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Fracture toughness in shales using nano-indentation

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ABSTRACT

Fracture toughness property is often ignored but has a significant impact on the hydraulic fracture design, i.e. length and height. It represents the resistance of the rock to hydraulic fracturing and is a direct indicator of rock fracability. It is difficult to measure fracture toughness in shales using current experimental procedures designed for conventional rocks largely due to mechanical and chemical instability in shales. This study measures fracture toughness in shales using nano-indentation. Nano-indentation can, in fact, be used to derive both tensile and shear dominated fracture toughness. The observations in the study show that while crack-length method derived fracture toughness is close to Mode I (tensile) fracture toughness, energy-method based fracture toughness is closer to Mode II (shear) fracture toughness.

Fracture toughness was calculated for 67 samples in different shale formations namely Marcellus, Wolfcamp, Woodford and Eagle Ford. The results show that mineralogy had a strong control on fracture toughness. Fracture toughness decreases with increasing porosity, TOC (Total Organic Carbon) and clay content. The fracture toughness also shows strong anisotropy with fracture toughness measured parallel to the bedding planes being generally 33% higher than the corresponding values measured normal to the bedding planes. The fracture toughness shows a direct correlation with other mechanical properties like Young's modulus, hardness and brittleness. Typically, samples with fracture toughness values greater than 3 MPa $m^{0.5}$ were found to be brittle based on Young's modulus and Poisson's ratio cross plots. Evaluation of fluid effects on fracture toughness shows that it decreases by 10–37% due to spontaneous imbibition with low salinity brines. This could help understand the impact of hydraulic fracturing fluid on fracture propagation and stimulated reservoir volume (SRV) creation.

1. Introduction

The operators working in unconventional shales are trying to understand fracture propagation and resulting stimulated reservoir. While some have tried the experimental approach i.e. conducting hydraulic fracture experiments in the lab, others have tried numerical modeling. The industry is also moving forward with field-level research experiments like hydraulic fracturing test site (HFTS) or Marcellus shale energy and environment laboratory (MSEEL). These different techniques are aimed at characterizing mechanical rock properties like fracture toughness which control fracture (or crack) propagation.

Crack is any opening in the rock with 1 or 2 dimensions much smaller than the third dimension. There are three different types of cracks based on the length of the crack (Engelder, 1987). Micro-cracks are the smallest with 1 to 10^4 μm in length, then meso-cracks (1–100 mm) and macro-cracks (0.1–100 m). Macrocracks are also termed as fractures. Fractures are formed when the stresses at the crack tip increase beyond a critical stress intensity factor which leads to crack propagation. The critical stress intensity factor is a characteristic mechanical rock property called fracture toughness. A newly created fracture typically has a main connected fracture but is also surrounded by a process zone where the cracks are still isolated (Hoagland et al., 1973; Atkinson et al., 1982; Backers, 2004; Ratzlaff, 2018).

Cracks can also be categorized based on the force that created them. Mode I cracks are created using tensile forces, Mode II and Mode III cracks are generated by shear forces (Engelder, 1987; Whittaker et al., 1992; Anderson, 2005). The difference between Mode II and Mode III is that the crack propagation is parallel to direction of force in Mode II and perpendicular in Mode III (Backers, 2004; Anderson, 2005). Any combination of the three modes is called mixed-mode fracturing. Fracture toughness values are also dependent on the mode of crack generation. In macroscale laboratory experiments, experimental design enables measurements of pure Mode I and Mode II fracture toughness values, but in indentation-based experiments using Atomic Force

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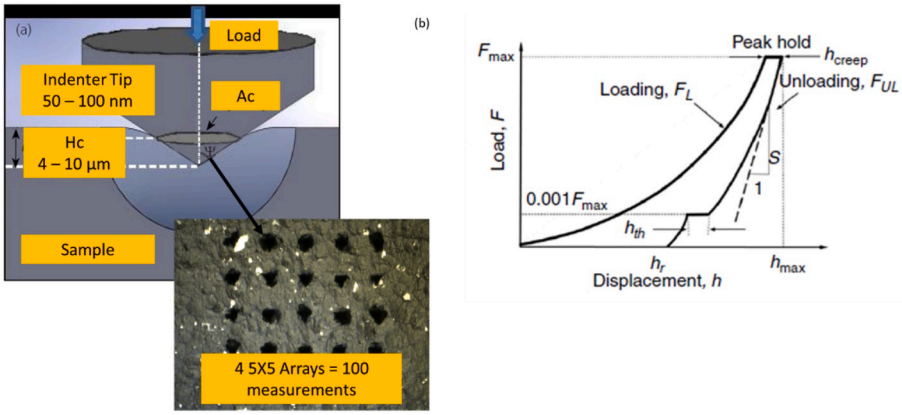


Fig. 1. Schematic of nano-indentation. (a) Shows the indentation on the sample surface with the indenter tip (Berkovich, 50–100 nm diameter). Measurements are made in 4 separate arrays of 25 measurements. Within each array, the indents are positioned to eliminate measurement interference. The sampling produces an average response which is less susceptible to sample heterogeneity. Indentations are made at 500 mN (maximum available) load to avoid measuring individual grains and maximizing the response volume. (b) Typical load-displacement profile of an individual indentation taken from Yang et al. (2018). The slope of the initial portion of the unloading curve (Stiffness, S) is used to calculate Young's modulus and hardness.

Microscopy/Nanoindenter, generally mixed mode fractures are created. These mixed-mode fractures can be either tensile or shear dominated. Field observations by Lockner et al. (1992) and Zang et al. (2000) suggest hydraulic fractures created in the subsurface are likely mixed mode fractures which are tensile dominated but have shear component as well. This suggests the possibility of obtaining field representative fracture toughness values using indentation-based methods.

Several laboratory testing methods have been proposed to get fracture toughness values for different modes. Mode I methods are most mature, reliable and yield consistent results. Some Mode II methods also exist but are not as reliable and very few methods are available to give pure Mode III fracture toughness (Cox and Scholz, 1988; Yacoub-Tokatly et al., 1989; Backers, 2004). ISRM (International Society of Rock Mechanics) suggests three methods to get Mode I fracture toughness namely Chevron Bend, Short Rod and Cracked Chevron Notched Brazilian Disc (Ouchterlony, 1988; Fowell, 1995). These methods use different geometrical configurations with tensile forces applied to the notch to measure Mode I fracture toughness. The notch is created in the center of the sample of known length and the applied load at which failure happens can be used to calculate the fracture toughness. Methods to obtain pure Mode II fracture toughness cannot completely avoid tensile stresses but the ones most commonly used are Four Point Bending Cube (Barr and Derradj, 1990), Punch through Shear (Watkins, 1983) and compression-shear cube test (Jumikis, 1979).

Clifton et al. (1976) measured fracture toughness on a set of sandstone (0.57–1.46 MPa $m^{0.5}$), siltstone (1.04–1.37 MPa $m^{0.5}$) and shale samples (0.61–1.29 MPa $m^{0.5}$). They also measured fracture surface roughness and found that the roughness was much higher in shale samples (3.81 mm) compared to sandstone or siltstone samples (1.27 mm) concluding fractures in shales follow more tortuous paths. Senseny and Pfeifle (1984) measured fracture toughness in 56 sandstone, shale, and mudstone samples from three wells in Piceance basin. The average value for sandstones was 1.27 MPa $m^{0.5}$ while for shales, mudstones and siltstones, the average value was 1.46 MPa $m^{0.5}$. Chavez et al. (2015) used notched Brazilian disc and semicircular three-point load tests to determine fracture toughness for Vaca Muerta shale samples (0.18–1.25 MPa $m^{0.5}$). Dudley et al. (2016) used cracked chevron notched Brazilian disc (CCNBD) method to measure fracture toughness for liquid-rich shales and developed correlations between fracture toughness and sonic velocities, density, gamma ray, etc. Suo et al. (2018) also used CCNBD method to measure fracture toughness in Longmaxi shale formation. They measured fracture toughness along different angles to the bedding plane to show the anisotropic nature of fracture toughness.

Nano-indentation allows accurate and repeatable measurements in localized areas at small loads making nano-indentation an ideal tool for measuring fracture toughness. Liu (2015) was the first one to measure fracture toughness in shales (Antrim and Opalinus) using nano-indentation. The calculated values of fracture toughness varied between 0.3 and 0.8 MPa $m^{0.5}$ in Antrim shale samples and between 0.7 and 1.5 MPa $m^{0.5}$ in Opalinus shale samples. Later, Liu et al. (2016) also reported fracture toughness measurements on Bakken samples using nano-indentation. The average value of calculated fracture toughness was 3 MPa $m^{0.5}$. They also published a correlation between Young's modulus and fracture toughness and found the two to be positively correlated. Yang et al. (2018) also measured fracture toughness on shale samples (1.5–2.8 MPa $m^{0.5}$) and carried out fluid sensitivity studies. They showed water-weakening as fracture toughness reduced up to 43% when dry samples were exposed to fresh water. They also showed that presence of hydrophobic coating on the sample surface could significantly reduce the water-weakening effects on fracture toughness.

1.1. Measurement procedures

The core plugs for this study were taken from four different shale plays namely Marcellus (17 depths), Wolfcamp (15 depths), Woodford (15 depths) and Eagle Ford (20 depths). Except Eagle Ford, only horizontal plugs were acquired. In Eagle Ford, both horizontal and vertical plugs were acquired at the 20 depths. The following petrophysical measurements were carried out for all the formations: porosity, TOC, mineralogy, ultrasonic velocities and nano-indentation. The plugs did not come from preserved cores; they were heated in an oven at 100 °C for 24 h before measurement.

TOC is measured using dry pyrolysis in LECO™ apparatus. The CO_2 released during the combustion of the organic matter is measured and converted to total organic carbon (TOC). A detailed procedure is available in Law (1999).

The porosity is measured on dried plugs using high-pressure (200 psi) helium porosimeter. First, mercury immersion technique is used to determine bulk volume of the sample. Next, the porosimeter uses Boyle's law concept to determine grain volume. The porosity is calculated using the formula:

$$\varnothing = \frac{V_{bulk} - V_{grain}}{V_{bulk}}$$

The high-pressure helium porosimetry cannot measure the pore volume occupied by residual fluids. Thus, to get the total porosity measurement, Nuclear Magnetic Resonance (NMR) is done on the same sample plug and the NMR porosity is added to the helium porosity to get the total porosity.

The mineralogy of the samples is determined using transmission Fourier Transform Infrared Spectroscopy (FTIR). The in-house algorithm identifies and quantifies (wt. %) 16 minerals commonly found in shales namely quartz, calcite, dolomite, aragonite, siderite, oligoclase, albite, orthoclase, illite, chlorite, kaolinite, smectite, mixed-layer clays, apatite, anhydrite and pyrite. The detailed procedure is given in Sondergeld and Rai (1993) and Ballard (2007).

Table 1
Parameters set in the nano-indenter to obtain load-displacement curves.

Surface Approach Distance (nm)	Surface Approach velocity (nm/s)	Maximum Load (mN)	Assumed Sample Poisson's ratio	Peak load hold time (s)	Time to load (s)
2000	20	500	0.25	10	15

The ultrasonic measurements were made using pulse transmission technique (Mattaboni and Schreiber, 1967). The sample is placed inside a neoprene jacket. The top and bottom faces of the sample are polished and made parallel to each other. Three piezoelectric transducers (P, S_1 and S_2) are placed on each face of the core plug to generate and record compressional (P) and two orthogonally polarized shear waves (S_1 and S_2). The length of the plug and travel times are used to determine velocities. Measurements are done at several confining pressure steps namely 500, 750, 1000, 1500, 2000, 2500, 3000, 4000, and 5000 psi. At higher pressures, the cracks tend to close and the measurements become representative of the in-situ state. Thus, the velocities at 5000 psi were used to calculate mechanical properties like Young's modulus and shear modulus (Birch, 1961). More details are available in Gupta et al. (2018).

An Agilent G200 nanoindenter is used for measuring Young's modulus, hardness and fracture toughness. The biggest advantage of nano-indentation is that it is essentially a non-destructive test and does not require large sample volumes (1–10 mm thick, 1 cm wide). The samples are manually polished and ion-milled using argon to make the sample flat and reduce the surface roughness to less than 0.1 μm. The sample preparation procedure is detailed in Kumar et al. (2012). Fig. 1 shows the schematic of a nano-indenter. The indenter tip moves using a magnet and coil assembly and the vertical displacement is measured using a capacitance displacement gauge. The key variables that control the indentation characteristics are summarized in Table 1.

In Fig. 1 (a), the projected area of contact A_c is calculated from the depth of contact (H_c) using equation (1). The constant 24.5 is dependent on the indenter tip.

$$A_c = 24.5H_c^2 \tag{1}$$

The early-time unloading curve in load-displacement curve is elastic and the slope S (Stiffness) can be used to determine both Young's modulus and hardness. Hardness (H) is determined using the maximum load (F_{max}) and area (A_c) in equation (2).

$$H = \frac{F_{max}}{A_c} \tag{2}$$

When the Berkovich tip indents the sample surface, both the indenter and the surface undergo deformation. Equation (3) can be used to calculate the reduced Young's modulus and equation (4) shows the relationship between the reduced Young's modulus (E_r), sample Young's modulus (E_s) and the indenter tip Young's modulus (E_i). The stiffness of the tip is much greater than the sample itself therefore the deformation of the tip is negligible compared to the sample.

$$E_r = \frac{1}{\beta}\frac{\sqrt{\pi}}{2}\frac{S}{\sqrt{A_c}} \tag{3}$$

$$\frac{1}{E_r} = \frac{1-v_s^2}{E_s} + \frac{1-v_i^2}{E_i} \tag{4}$$

β is a constant which depends on indenter geometry and its value is 1.034 for a Berkovich tip. To calculate the sample's Young's modulus using equation (4), an estimate of sample's Poisson's ratio (v_s) is needed. In this study, a default value of 0.25 was used for all the samples. Hay and Pharr (2000) have shown that a ±0.1 uncertainty in Poisson's ratio results in only about 5% uncertainty in Young's modulus. Considering the dynamic range of Poisson's ratio varies between 0 and 0.5, it is evident that Young's modulus value is not very sensitive to Poisson's ratio value.

Liu (2015) details two procedures to calculate fracture toughness using a nanoindenter. The first one generates a radial crack, the length of which is used to determine fracture toughness using Equation (5).

$$K_{IC} = \alpha\left[\frac{E}{H}\right]^{0.5}\left[\frac{F_{max}}{c^{1.5}}\right] \tag{5}$$

In the equation, α is an empirical constant which depends on the indenter geometry. For the cube corner indenter, its value is 0.032; while for Berkovich indenter, it is 0.016. E is the Young's modulus, H is hardness, F_{max} is the maximum indenter load and c is the average crack length. Cube corner is generally used for this method as it is very sharp and capable of creating cracks easily. However, as Fig. 2 shows, while this method works very well in fused silica, the method fails in shales. Several combinations of loading rate and maximum load were tried to induce fused-silica like cracks in shale samples, but reliable fracture toughness values could not be estimated using this method on shales.

The "Energy Approach" method to calculate fracture toughness in shales using nano-indentation is detailed in Yang et al. (2018). This method developed by Li et al. (1997) for thin films was first applied to shales by Liu (2015). The total energy (U_t) during the indentation process (loading and unloading) is a sum of the elastic (U_e), pure plastic (U_{pp}) and fracture (U_{frac}) energies. Both pure plastic and fracture energies represent the irreversible work produced in the system.

$$U_t = U_e + U_{pp} + U_{frac} \tag{6}$$

The total energy can be calculated using equation (7) where F_L represents the loading force, h_{max} is the maximum displacement and

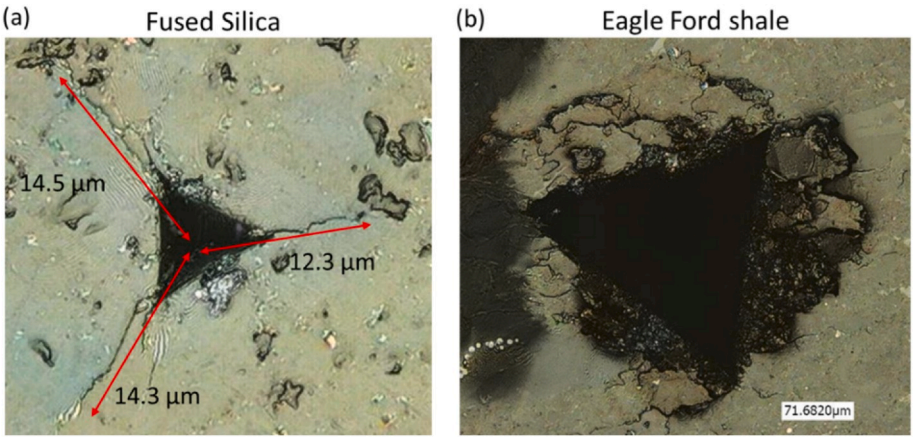


Fig. 2. Schematic of radial crack length measurement method for calculating fracture toughness. (a) shows the indentation on fused silica (using corner cube tip) which clearly shows cracks emanating from indentation tips (highlighted by red arrows). (b) shows the indentation (using corner cube tip) on Eagle Ford shale. Shale samples do not have simple cracks rather have a crack network making it difficult to measure crack length and estimate fracture toughness. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

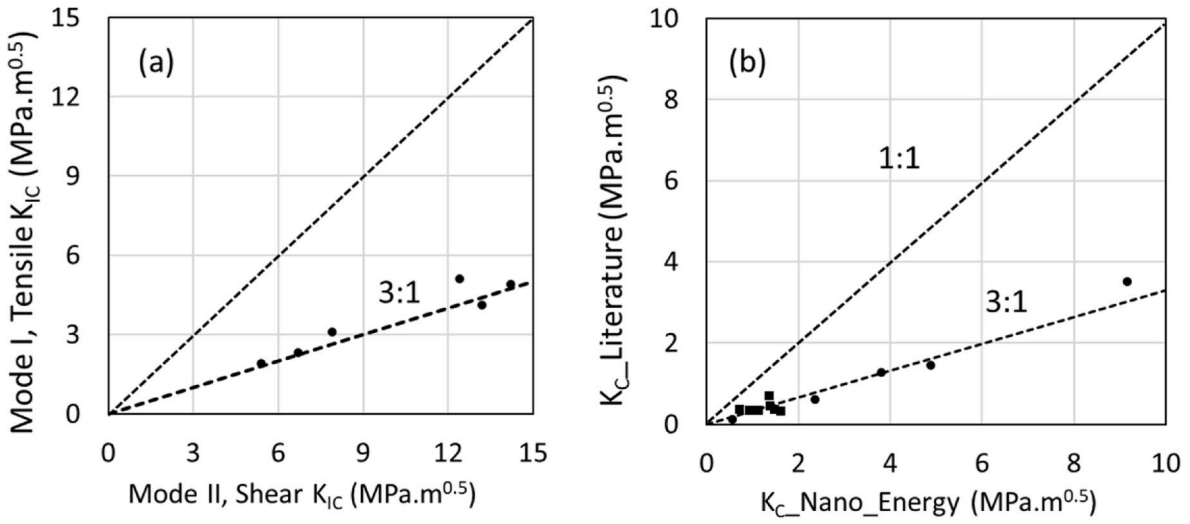


Fig. 3. (a) Comparison of Mode I and Mode II fracture toughness in limestone, granite, sandstone and diorite rocks. Mode II fracture toughness is roughly three times Mode I fracture toughness. (b) Mode I literature reported fracture toughness values compared with fracture toughness estimated using the nano-indentation energy method. It shows a 3:1 ratio similar to the one derived from Backers (2004) for Mode I and Mode II fracture toughness values. The data for Opalinus shale (black square points) from Liu (2015) is also plotted. The data corresponds to fracture toughness derived using crack length method (y-axis) and energy method (x-axis).

h_{creep} is the creep displacement that occurs while holding the maximum constant load (Fig. 1).

$$U_t = \int_0^{h_{max}-h_{creep}} F_L dh \tag{7}$$

Elastic energy (U_e) and pure plastic energy (U_{pp}) can be calculated by equations (8) and (9), respectively. F_{UL} stands for unloading force, h_r is the residual displacement, F_{max} is the maximum load, h_{th} is the thermal drift of the system.

$$U_e = \int_{h_r}^{h_{max}} F_{UL} dh - 0.01 F_{max} h_{th} \tag{8}$$

$$U_{pp} = \left\{ 1 - \left[\frac{1 - 3\left(\frac{h_f}{h_{max}}\right)^2 + 2\left(\frac{h_f}{h_{max}}\right)^3}{1 - \left(\frac{h_f}{h_{max}}\right)^2} \right] \right\} U_t \tag{9}$$

$$h_f = h_r - h_{th} - h_{creep} \tag{10}$$

Equations (11) and (12) are used to estimate fracture toughness from the fracture energy. G_c is the strain energy release rate and it represents the energy dissipated from fracture per unit area of the newly created fracture face.

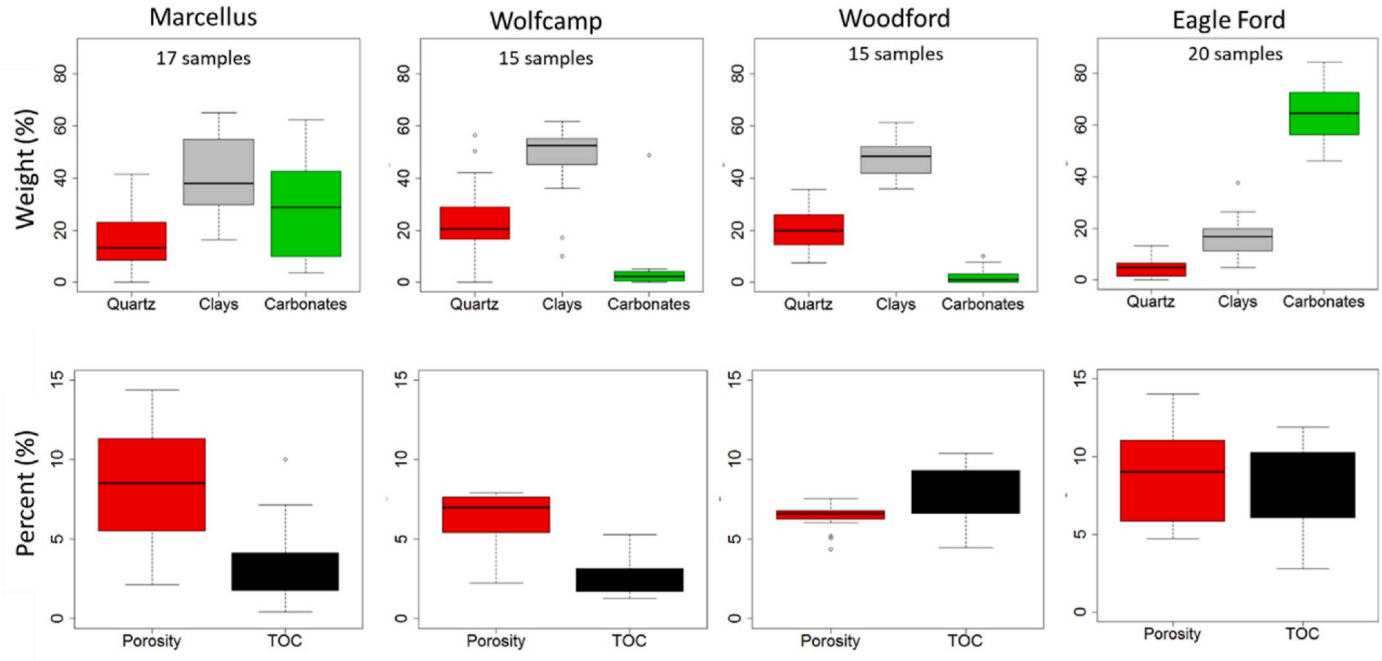


Fig. 4. The boxplots show the mineralogy and petrophysical properties of the samples. It is evident that Wolfcamp and Woodford have high clay content while Eagle Ford and Marcellus have high carbonate content. The TOC and porosity box plots show that Eagle Ford and Marcellus have high porosities while Woodford and Eagle Ford have higher TOC.

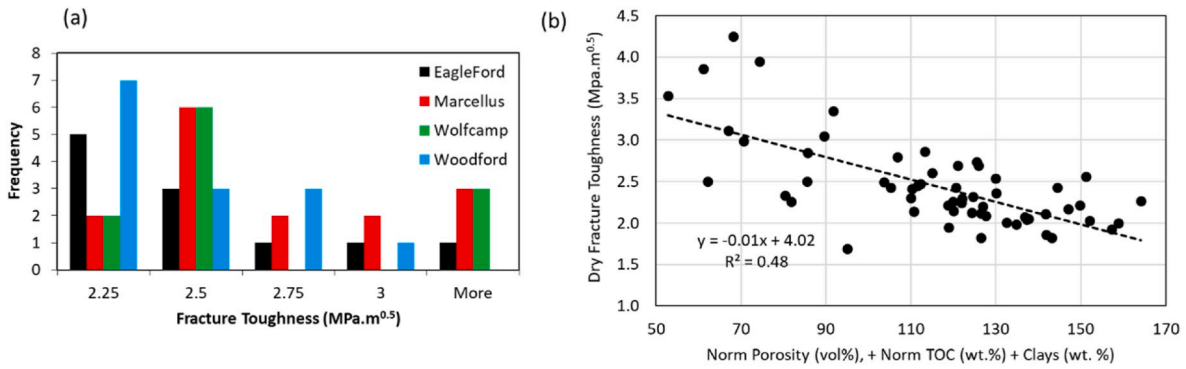


Fig. 5. (a) The histogram of the fracture toughness values for horizontal plugs from different formations. Eagle Ford and Woodford have lower values of fracture toughness shown by higher frequency at lower values. (b) Fracture toughness as a function of porosity, TOC and clay content. Increase in the percent of soft components reduces the fracture toughness.

$$G_c = \frac{\partial U_{frac}}{\partial A_c} = \frac{U_{frac}}{A_c} \tag{11}$$

$$K_C = \sqrt{G_c E_r} \tag{12}$$

In the equation, E_r is the reduced Young's modulus and A_c is the projected contact area. This method was used to estimate the fracture toughness in this study.

2. Results and discussion

The motivation behind measuring fracture toughness is that it has a big impact on the design of hydraulic fractures and their resultant geometries. Dudley et al. (2016) showed that fracture propagation is affected by fracture toughness of the different zones, with fracture growth halted by zones of higher fracture toughness. Thus, fracture toughness can play an important role in determining fracture height. This is especially important for deciding well spacing and placement in stacked plays.

Backers (2004) reported both Mode I and Mode II fracture toughness on samples in limestone, granite, sandstone and diorite. Fig. 3 (a) shows that Mode II fracture toughness is roughly three times Mode I fracture toughness. To further understand fracture modes, few standards were chosen, and their literature reported Mode I fracture toughness values were compared with nano-indentation "energy-method" derived fracture toughness values (see Fig. 3 (b)). The Mode I fracture toughness values were taken from following references – alumina (Accuratus, 2018), Na-Montmorillonite (Vanorio et al., 2003), Lyon sandstone (Senseny and Pfeifle, 1984), pyrophyllite (Hantal et al., 2014) and soda lime glass (Wright, 2010). The comparison shows that the nano-indentation energy method-based fracture toughness values are roughly three times Mode I fracture toughness, suggesting they closely approximate Mode II fracture toughness.

The data for Opalinus shale from Liu (2015) is also plotted in Fig. 3 (b). The data corresponds to fracture toughness derived using crack length method (y-axis) and energy method (x-axis). The data also lies on 3:1 line suggesting that crack-length based fracture toughness values are tensile dominated and more closely approximate Mode I fracture toughness.

The fracture toughness was determined for several samples (67 depths) across four different shale plays namely Marcellus, Wolfcamp, Woodford and Eagle Ford. Fig. 4 shows the petrophysical properties for the different samples. It is evident that Wolfcamp and Woodford have high clay content while Eagle Ford and Marcellus have high carbonate content. The TOC and porosity box plots show that Eagle Ford and Marcellus have high porosities while Woodford and Eagle Ford have higher TOC.

Fig. 5(a) shows the histogram of the fracture toughness values for the different formations measured using nano-indentation energy method. Eagle Ford and Woodford have lower values of fracture toughness shown by higher frequency at lower values. This may be due to higher TOC in both Woodford and Eagle Ford, higher porosity in Eagle Ford and higher clays in Woodford i.e. higher fraction of soft components. The fracture toughness values are plotted as a function of porosity, TOC and clay content in Fig. 5(b). It is evident that increasing TOC, clay and porosity content decreases the fracture toughness. The porosity (vol%) and TOC (wt.%) values are much smaller than clay content (wt.%) which may reduce the sensitivity of fracture toughness to porosity and TOC, when all the soft components are summed. Thus, porosity and TOC were normalized (equation (13)) before being plotted in Fig. 5(b).

$$Var_{norm} = \frac{Var - Var_{min}}{Var_{max} - Var_{min}} * 100 \tag{13}$$

Since clay minerals and TOC often cause anisotropy in shales, and fracture toughness is a strong function of TOC and clay content, it is likely that fracture toughness is anisotropic. Suo et al. (2018) and Chandler et al. (2016) showed that fracture toughness property is anisotropic. In Eagle Ford shale, both horizontal and vertical samples

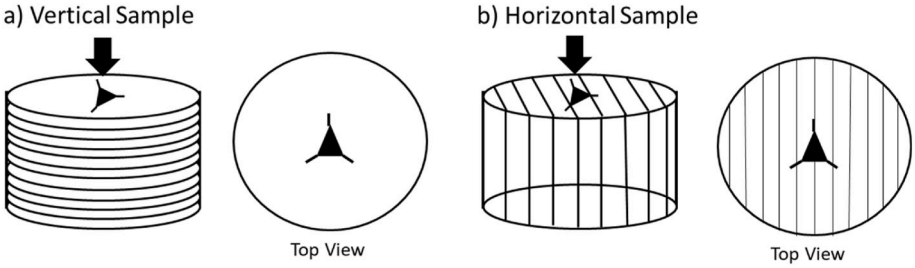


Fig. 6. Schematic of a) vertical sample and b) horizontal sample showing the position of nano-indentation with respect to the bedding planes. The indentations are generally 2–5 μm in depth and the lateral extension varies between 20 and 40 μm. The nano-indentation in a horizontal sample has a higher probability of cutting across different bedding planes compared to a vertical sample.

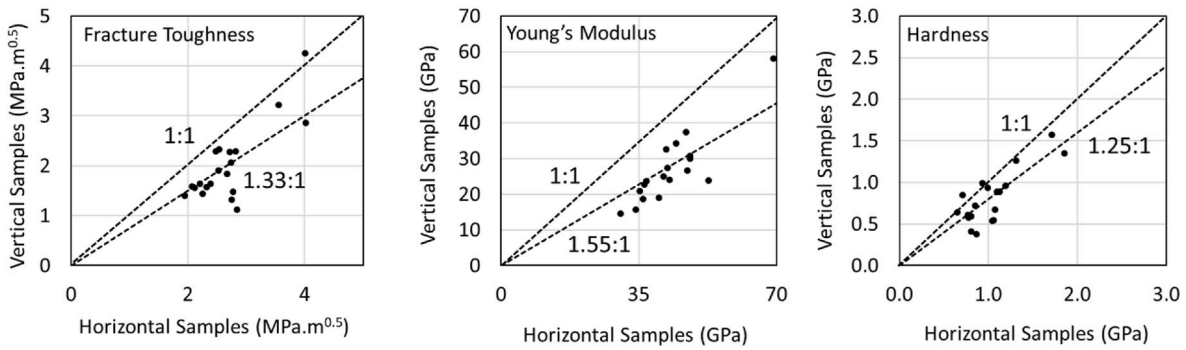


Fig. 7. Comparison of mechanical properties like fracture toughness, Young's modulus and hardness, of horizontal and vertical samples measured using nano-indentation. The plot suggests that fracture toughness measured on horizontal samples is on average 33% higher than the corresponding values measured on vertical samples. The trend is consistent with other mechanical properties like Young's modulus and hardness where the measurements on horizontal samples are 55% and 25% higher, respectively.

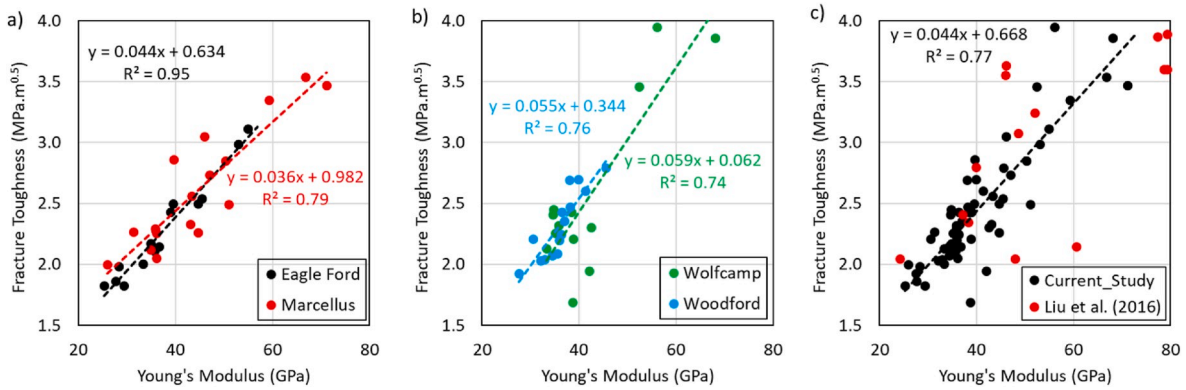


Fig. 8. Correlation between Young's modulus and fracture toughness both derived from nano-indentation on horizontal plugs. There is a direct correlation which suggests that fracture toughness can be estimated from Young's modulus. It is likely because Young's modulus is an indicator of fracture strength and higher strength implies higher resistance to fracture which results in higher fracture toughness.

were available. The indentations are generally 2–5 μm in depth and the lateral extension varies between 20 and 40 μm. In horizontal samples, the direction of indentation is parallel to the bedding plane while in vertical samples, the direction of indentation is perpendicular to the bedding planes (Fig. 6). The nano-indentation in a horizontal sample has a higher probability of cutting across several bedding planes compared to a vertical sample due to much higher lateral dimensions (20–40 μm) of the indentations (Fig. 6(b)). On the other hand, indentation in a vertical sample has a much lower probability of cutting across several bedding planes due to lower vertical dimensions (2–5 μm) of the indentations (Fig. 6(a)). Fig. 7 shows the comparison of several mechanical properties between horizontal and vertical samples all measured through nano-indentation. The plot suggests that fracture toughness measured on horizontal samples is on average 33% higher than the corresponding values measured on the vertical samples. This is consistent with earlier studies like Chandler et al. (2016) which show that fracture toughness measured when cracks are cutting across several bedding planes is higher compared to when cracks are propagating within a layer. The trend is also consistent with other mechanical properties like Young's modulus and hardness where the measurements on horizontal samples are 55% and 25% higher, respectively.

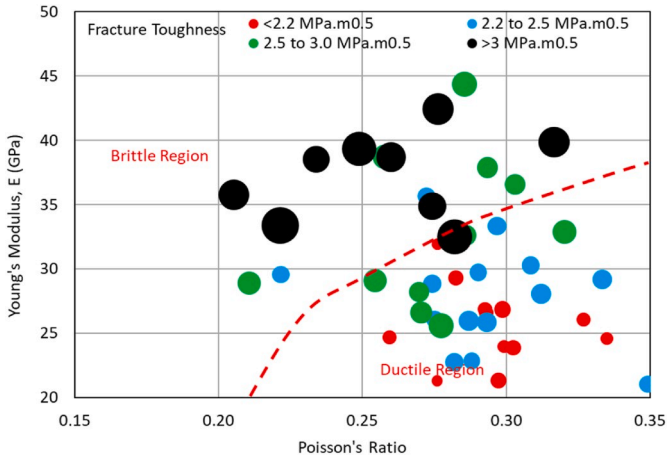


Fig. 9. Cross plot of Young's modulus and Poisson's ratio where the data points are sized based on fracture toughness. The Young's modulus and Poisson's ratio were calculated from ultrasonic (V_p, Vs) measurements independently from nano-indentation carried out on twin samples. It is evident that larger values of fracture toughness are associated with brittle rocks.

Fig. 8 shows the correlation between Young's modulus and fracture toughness both derived from nano-indentation. This study and previous studies (Liu et al., 2016) show there is a general trend that fracture toughness increases with Young's modulus. It is likely because Young's modulus is an indicator of fracture strength and higher strength implies higher resistance to fracture which results in higher fracture toughness. Fig. 8(a and b) show the correlations for the four formations in this study and they show similar trends with subtle variations among them. Fig. 8 (c) shows the data for all the four formations together. The data from Bakken (Liu et al., 2016) is also plotted and it follows the same trend but has more scatter. This universal correlation can be used when formation-specific correlation is not available. A direct correlation

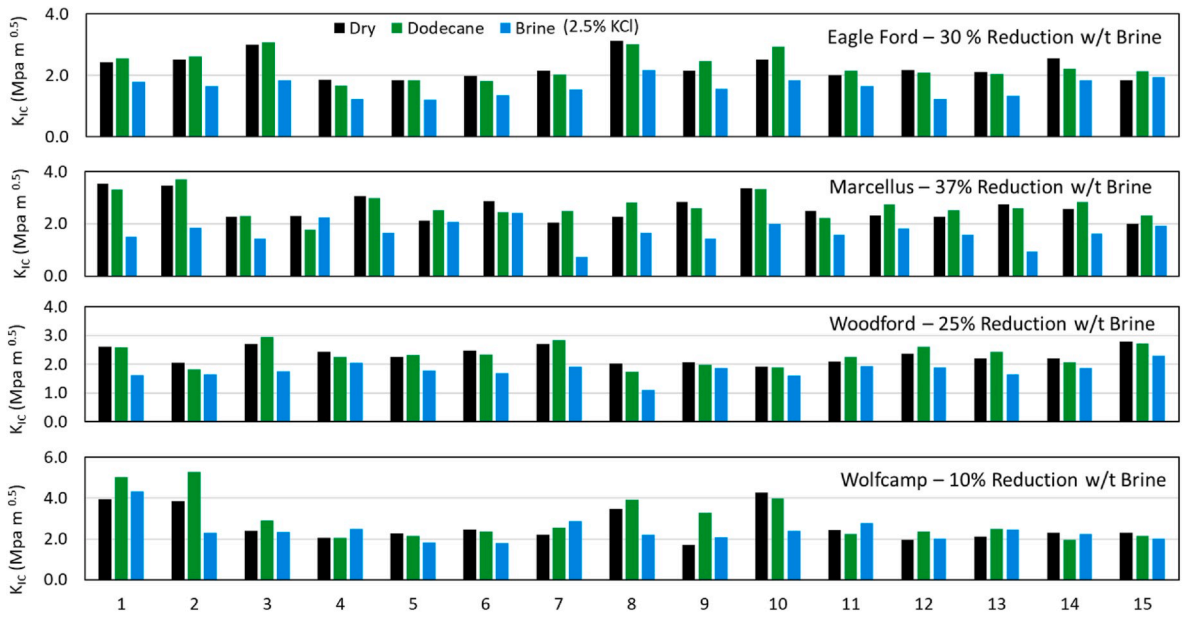


Fig. 10. The fracture toughness values for different samples in dry, dodecane and brine imbibed conditions. Several previous studies have shown that fluids weaken the rock and strongly impact their mechanical properties. The results in this study show that fracture toughness reduces on average between 10 and 37% for different shale plays on exposure to 2.5% KCl brine.

between the two is important as Young's modulus can be calculated from sonic. The fracture toughness on the other hand is tough to calculate from logs and values are sparsely published in the literature especially for shales.

Fracture toughness is a direct measure of rock fracability. This study shows fracture toughness to be an indirect indicator of rock brittleness, as well. The Grieser and Bray (2007) diagram commonly used in the industry shows the brittle and ductile regions separated on a Young's modulus and Poisson's ratio cross plot. For all the samples, the ultrasonic velocity and density measurements were also made. Using the Birch (1961) equations, the values of Young's modulus and Poisson's ratio were calculated. The Birch formulae are applicable to isotropic materials. However, for simplicity, but they were applied to shales in this study. The data was plotted on the same cross plot and brittle and ductile samples were identified (Fig. 9). In Fig. 9, it is evident that larger values of fracture toughness are associated with brittle rocks. Typically, samples having a fracture toughness value of 3 MPa $m^{0.5}$ or more lie in the brittle region.

Lastly, we looked at the fluid effects on fracture toughness. Several previous studies (Wick, 2015; Gupta et al. 2017, 2019) have shown fluid exposure can weaken the rock and strongly impact their mechanical properties. In this study, the effects of dodecane and 2.5% KCl brine were evaluated on fracture toughness. Duplicates were prepared for the 62 samples and were exposed to dodecane and 2.5% KCl brine. The samples were spontaneously imbibed with different fluids for 60 h. The detailed procedure for sample preparation and imbibition is given in Gupta et al. (2019). Once imbibed, nano-indentation measurements were carried out on the samples. The samples were mounted in a specialized holder which presented a fluid free surface to the indenter tip but had a cavity around the sample to hold the fluid while indenting. This was done to compensate for evaporation losses from the sample during indentation (~9 h per sample). The fracture toughness values were estimated from the load-displacement curves. The values for the dry, dodecane imbibed and brine imbibed samples are shown in Fig. 10. The dodecane imbibition had little to no effect on the fracture toughness but brine exposure significantly weakened the rock, leading to lower fracture toughness values. Out of the total 62 samples, 54 samples showed weakening with brine. Eagle Ford on average showed 30% reduction in fracture toughness, Marcellus showed 37% reduction, Woodford and Wolfcamp showed 25% and 10% reductions, respectively. Reduction in fracture toughness suggests rock weakening and it should fracture more easily, suggesting a lower breakdown pressure. Patel et al. (2017) compared breakdown pressures on Tennessee sandstone in dry and saturated states. He used 5.5″ long and 4" wide samples in the triaxial pressure setup. The breakdown pressure decreased from 3000 psi to 2100 psi, suggesting about 30% reduction in breakdown pressure due to exposure to brine. These results are consistent with the fracture toughness water weakening results seen in this study.

3. Conclusions

The study measures fracture toughness using nano-indentation on several samples from four different shale plays namely Woodford, Eagle Ford, Wolfcamp and Marcellus. The energy-approach is used to calculate the fracture toughness. The crack-length measurement method could not be applied as cracks generated using corner cube tip formed a complex network as opposed to singular-measurable tip-crack. The data suggests that the fracture toughness determined using the energy method is more shear dominated and closely approximate the Mode II fracture toughness. As opposed to this, the fracture toughness values measured using the crack length method closely approximate the Mode I tensile fracture toughness. Our study shows that fracture toughness is strongly dependent on mineralogy and is anisotropic; on average 33% higher parallel to the bedding plane as compared to when measured normal to the bedding. While fracture toughness is a direct indicator of rock fracability, the study also showed it to be an indirect indicator of brittleness. Samples with higher fracture toughness are found to be more brittle based on Young's modulus and Poisson's ratio cross plot. Lastly, fracture toughness like other mechanical properties, also exhibits water-weakening. On average, the values of fracture toughness reduce by 10–37% for the different shale plays.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.petrol.2020.107222.

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